Setup Guide

HBase

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Database

Contents

HBase 3-Node Cluster

Complete Setup & Configuration Guide

VMware Virtual Machines | Master + Backup Master + 2 Region Servers

Cluster Node Overview

Hostname	IP Address	Role
master	192.168.1.100	HBase Master (Primary) + NameNode + ZooKeeper
rs1	192.168.1.101	Region Server 1 + Backup Master + DataNode
rs2	192.168.1.102	Region Server 2 + DataNode

📝 **Note:** Adjust IP addresses to match your VMware network subnet. These are examples.

Section 1: Prerequisites & VMware VM Setup

1.1 Software Requirements

Component	Version	Notes
Operating System	Ubuntu 20.04 LTS / CentOS 7+	64-bit recommended
Java (JDK)	OpenJDK 8 or 11	HBase requires Java 8+
Hadoop	3.3.x	HDFS for HBase storage
HBase	2.4.x	Latest stable release
ZooKeeper	3.5.x	Bundled with HBase

1.2 VMware VM Specifications (Each Node)

Resource	Minimum	Recommended
vCPU	2 cores	4 cores
RAM	4 GB	8 GB
Disk	40 GB	100 GB
Network Adapter	NAT or Host-Only	Bridged (for inter-VM communication)

> ⚠ **Important:** Set all 3 VMs to Bridged network mode in VMware so they can communicate with each other and get unique IPs.

1.3 Set Static IP Addresses on Each VM

On each VM, configure a static IP. Example for Ubuntu (edit /etc/netplan/00-installer-config.yaml):

```
network:
  version: 2
  ethernets:
    ens33:                          # Replace with your interface name
      dhcp4: no
      addresses: [192.168.1.100/24]  # Use .101 for rs1, .102 for rs2
      gateway4: 192.168.1.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Apply the configuration:

```
sudo netplan apply
```

1.4 Configure /etc/hosts on ALL Three Nodes

Run this on master, rs1, and rs2:

```
sudo nano /etc/hosts
```

Add these lines (replace with your actual IPs):

```
192.168.1.100   master
192.168.1.101   rs1
192.168.1.102   rs2
```

1.5 Set Hostnames

On master VM:

```
sudo hostnamectl set-hostname master
```

On rs1 VM:

```
sudo hostnamectl set-hostname rs1
```

On rs2 VM:

```
sudo hostnamectl set-hostname rs2
```

Section 2: User & SSH Setup (All Nodes)

2.1 Create hadoop User on ALL Nodes

Run on master, rs1, and rs2:

```
sudo adduser hadoop
sudo usermod -aG sudo hadoop
su - hadoop
```

2.2 Generate SSH Key on Master

Run only on master node as hadoop user:

```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

2.3 Copy SSH Keys to rs1 and rs2

From the master node, copy the key to both region servers:

```
ssh-copy-id hadoop@rs1
ssh-copy-id hadoop@rs2
```

Test passwordless SSH (should connect without password prompt):

```
ssh hadoop@rs1
ssh hadoop@rs2
```

2.4 Copy SSH Keys on rs1 (for Backup Master)

Since rs1 is also backup master, it needs SSH access to master and rs2:

```
# On rs1 node
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
ssh-copy-id hadoop@master
ssh-copy-id hadoop@rs2
```

Section 3: Java Installation (All Nodes)

3.1 Install OpenJDK 8

Run on master, rs1, and rs2:

```
sudo apt update
sudo apt install -y openjdk-8-jdk
```

3.2 Verify Java Installation

```
java -version
```

Expected output: openjdk version "1.8.0_xxx"

3.3 Set JAVA_HOME Environment Variable

Find Java path:

```
readlink -f $(which java)
```

Add to ~/.bashrc on ALL nodes (as hadoop user):

```
echo 'export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64' >> ~/.bashrc
echo 'export PATH=$PATH:$JAVA_HOME/bin' >> ~/.bashrc
source ~/.bashrc
```

Section 4: Hadoop Installation & Configuration

4.1 Download & Install Hadoop (All Nodes)

Run on master, rs1, and rs2:

```
cd /opt
sudo wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
sudo tar -xzf hadoop-3.3.6.tar.gz
sudo mv hadoop-3.3.6 hadoop
sudo chown -R hadoop:hadoop /opt/hadoop
```

4.2 Set Hadoop Environment Variables (All Nodes)

Add to ~/.bashrc on ALL nodes:

```
echo 'export HADOOP_HOME=/opt/hadoop' >> ~/.bashrc
echo 'export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop' >> ~/.bashrc
echo 'export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin' >> ~/.bashrc
source ~/.bashrc
```

4.3 Configure hadoop-env.sh (All Nodes)

```
sudo nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh
```

Find and set JAVA_HOME line:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

4.4 Configure core-site.xml (All Nodes)

```
sudo nano $HADOOP_HOME/etc/hadoop/core-site.xml
```

Set content to:

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://master:9000</value>
  </property>
  <property>
    <name>hadoop.tmp.dir</name>
    <value>/opt/hadoop/tmp</value>
  </property>
</configuration>
```

Create the tmp directory on all nodes:

```
sudo mkdir -p /opt/hadoop/tmp
sudo chown -R hadoop:hadoop /opt/hadoop/tmp
```

4.5 Configure hdfs-site.xml (All Nodes)

```
sudo nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

Set content to:

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>2</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>/opt/hadoop/data/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>/opt/hadoop/data/datanode</value>
  </property>
</configuration>
```

Create data directories on all nodes:

```
sudo mkdir -p /opt/hadoop/data/namenode
sudo mkdir -p /opt/hadoop/data/datanode
sudo chown -R hadoop:hadoop /opt/hadoop/data
```

4.6 Configure workers File (Master Node Only)

```
sudo nano $HADOOP_HOME/etc/hadoop/workers
```

Replace contents with:

```
rs1
rs2
```

4.7 Format HDFS NameNode (Master Node Only)

```
hdfs namenode -format
```

⚠ **Important:** Only run format once! Re-formatting will erase all HDFS data.

4.8 Start Hadoop Cluster (Master Node)

```
start-dfs.sh
```

Verify Hadoop is running using jps on each node:

On master — should show: NameNode, SecondaryNameNode

On rs1 and rs2 — should show: DataNode

Section 5: HBase Installation & Configuration

5.1 Download & Install HBase (All Nodes)

Run on master, rs1, and rs2:

```
cd /opt
sudo wget https://downloads.apache.org/hbase/2.4.17/hbase-2.4.17-bin.tar.gz
sudo tar -xzf hbase-2.4.17-bin.tar.gz
sudo mv hbase-2.4.17 hbase
sudo chown -R hadoop:hadoop /opt/hbase
```

5.2 Set HBase Environment Variables (All Nodes)

Add to ~/.bashrc on ALL nodes:

```
echo 'export HBASE_HOME=/opt/hbase' >> ~/.bashrc
echo 'export PATH=$PATH:$HBASE_HOME/bin' >> ~/.bashrc
source ~/.bashrc
```

5.3 Configure hbase-env.sh (All Nodes)

```
sudo nano $HBASE_HOME/conf/hbase-env.sh
```

Set/uncomment these lines:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HBASE_MANAGES_ZK=true
export HBASE_HOME=/opt/hbase
```

5.4 Configure hbase-site.xml (All Nodes)

```
sudo nano $HBASE_HOME/conf/hbase-site.xml
```

Set content to:

```
<configuration>
  <property>
    <name>hbase.rootdir</name>
    <value>hdfs://master:9000/hbase</value>
  </property>
  <property>
    <name>hbase.cluster.distributed</name>
    <value>true</value>
  </property>
  <property>
    <name>hbase.zookeeper.quorum</name>
    <value>master,rs1,rs2</value>
  </property>
```

```
  <property>
    <name>hbase.zookeeper.property.dataDir</name>
    <value>/opt/hbase/zookeeper</value>
  </property>
  <property>
    <name>hbase.zookeeper.property.clientPort</name>
    <value>2181</value>
  </property>
  <property>
    <name>hbase.master</name>
    <value>master:16000</value>
  </property>
  <property>
    <name>hbase.regionserver.port</name>
    <value>16020</value>
  </property>
  <property>
    <name>hbase.master.info.port</name>
    <value>16010</value>
  </property>
</configuration>
```

5.5 Configure regionservers File (Master Node Only)

```
sudo nano $HBASE_HOME/conf/regionservers
```

Replace contents with:

```
rs1
rs2
```

5.6 Configure backup-masters File (Master Node Only)

This file tells HBase which node is the backup master:

```
sudo nano $HBASE_HOME/conf/backup-masters
```

Add this single line:

```
rs1
```

📝 **Note:** This makes rs1 the backup master. If the primary master fails, rs1 can take over.

5.7 Create ZooKeeper Data Directory (All Nodes)

```
sudo mkdir -p /opt/hbase/zookeeper
sudo chown -R hadoop:hadoop /opt/hbase/zookeeper
```

5.8 Distribute Configurations to rs1 and rs2

From master, copy config files to region servers:

```
scp $HBASE_HOME/conf/hbase-site.xml hadoop@rs1:$HBASE_HOME/conf/
scp $HBASE_HOME/conf/hbase-env.sh  hadoop@rs1:$HBASE_HOME/conf/
scp $HBASE_HOME/conf/hbase-site.xml hadoop@rs2:$HBASE_HOME/conf/
scp $HBASE_HOME/conf/hbase-env.sh  hadoop@rs2:$HBASE_HOME/conf/
```

Section 6: Firewall & Port Configuration

6.1 Open Required Ports (All Nodes)

If UFW is enabled, open the required ports:

```
sudo ufw allow 2181   # ZooKeeper client port
sudo ufw allow 2888   # ZooKeeper peer port
sudo ufw allow 3888   # ZooKeeper leader election
sudo ufw allow 9000   # HDFS NameNode RPC
sudo ufw allow 9870   # HDFS Web UI
sudo ufw allow 16000  # HBase Master RPC
sudo ufw allow 16010  # HBase Master Web UI
sudo ufw allow 16020  # HBase RegionServer RPC
sudo ufw allow 16030  # HBase RegionServer Web UI
```

Port	Service	Used By
2181	ZooKeeper Client	All nodes
2888	ZooKeeper Peer	ZooKeeper quorum
3888	ZooKeeper Election	ZooKeeper quorum
9000	HDFS NameNode RPC	master
9870	HDFS Web UI	master
16000	HBase Master RPC	master
16010	HBase Master Web UI	master
16020	HBase RegionServer RPC	rs1, rs2
16030	HBase RegionServer Web UI	rs1, rs2

Section 7: Starting the HBase Cluster

7.1 Ensure Hadoop is Running First

From master node:

```
start-dfs.sh
jps   # Verify NameNode is running
```

7.2 Start HBase (Master Node Only)

```
start-hbase.sh
```

This single command starts HBase on all nodes including region servers and backup master.

7.3 Verify Processes with jps

Node	Expected jps Output
master	HMaster, HQuorumPeer, NameNode, SecondaryNameNode, Jps
rs1	HRegionServer, HMaster (backup), HQuorumPeer, DataNode, Jps
rs2	HRegionServer, HQuorumPeer, DataNode, Jps

7.4 Access Web UIs

Web Interface	URL	Description
HBase Master UI	http://master:16010	Cluster overview, region info
HDFS NameNode UI	http://master:9870	File system health
RS1 Region UI	http://rs1:16030	Region server details
RS2 Region UI	http://rs2:16030	Region server details

Section 8: Testing the Cluster

8.1 Open HBase Shell

```
hbase shell
```

8.2 Run Basic HBase Commands

Check cluster status:

```
hbase> status
hbase> status 'detailed'
```

Create a test table:

```
hbase> create 'test_table', 'cf1'
```

Insert test data:

```
hbase> put 'test_table', 'row1', 'cf1:name', 'Alice'
hbase> put 'test_table', 'row2', 'cf1:name', 'Bob'
```

Read data back:

```
hbase> scan 'test_table'
hbase> get 'test_table', 'row1'
```

List all tables:

```
hbase> list
```

Exit the shell:

```
hbase> exit
```

8.3 Verify ZooKeeper Quorum

```
echo ruok | nc master 2181
echo ruok | nc rs1 2181
echo ruok | nc rs2 2181
```

Each command should return: imok

Section 9: Stopping the Cluster

9.1 Stop HBase (Master Node)

```
stop-hbase.sh
```

9.2 Stop Hadoop (Master Node)

```
stop-dfs.sh
```

Section 10: Troubleshooting

10.1 Common Issues

Issue	Cause	Fix
RegionServer not joining	SSH/hostname issue	Verify /etc/hosts and passwordless SSH
ZooKeeper not starting	Port conflict or dir missing	Check port 2181 and zookeeper dataDir
HDFS not accessible	NameNode not formatted	Run: hdfs namenode -format
HMaster fails to start	Java not found	Verify JAVA_HOME in hbase-env.sh
Connection refused on Web UI	Firewall blocking	Open ports with ufw allow
Backup master not showing	backup-masters file missing	Create $HBASE_HOME/conf/backup-masters with rs1

10.2 Useful Log Locations

```
# HBase logs
tail -f $HBASE_HOME/logs/hbase-hadoop-master-master.log
tail -f $HBASE_HOME/logs/hbase-hadoop-regionserver-rs1.log
```

```
# HDFS logs
tail -f $HADOOP_HOME/logs/hadoop-hadoop-namenode-master.log
```

10.3 Restart Individual Services

```
# Restart a specific RegionServer (from master)
$HBASE_HOME/bin/hbase-daemon.sh stop regionserver   # on rs1/rs2
$HBASE_HOME/bin/hbase-daemon.sh start regionserver  # on rs1/rs2
```

```
# Restart backup master on rs1
$HBASE_HOME/bin/hbase-daemon.sh stop master   # on rs1
$HBASE_HOME/bin/hbase-daemon.sh start master  # on rs1
```

Quick Reference: Configuration File Summary

File	Location	Configured On
hadoop-env.sh	$HADOOP_HOME/etc/hadoop/	All nodes
core-site.xml	$HADOOP_HOME/etc/hadoop/	All nodes
hdfs-site.xml	$HADOOP_HOME/etc/hadoop/	All nodes
workers	$HADOOP_HOME/etc/hadoop/	Master only
hbase-env.sh	$HBASE_HOME/conf/	All nodes
hbase-site.xml	$HBASE_HOME/conf/	All nodes
regionservers	$HBASE_HOME/conf/	Master only
backup-masters	$HBASE_HOME/conf/	Master only

Document prepared for a 3-node HBase cluster on VMware virtual machines.